REPRODUCIBILITY_STATEMENT.md

Deliverable D-S1 — Reproducibility statement
Project: AMGCR Earthquake Research
Scope: California IRIS / EarthScope FDSN pilot (acquisition → feature matrix)
Date: 29 July 2026

This document describes how to reproduce the **completed certificate-track California pilot workflow** documented in the repository. It does **not** cover EarthESND K-NET scientific reproduction or future European pilots.

Related:
DATA_AVAILABILITY_STATEMENT.md · IRIS_DATASET_REPORT.md · FINAL_DELIVERABLE_AUDIT.md

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1. What this workflow reproduces

The pilot chain implemented in `scripts/download/` and `scripts/analysis/` produces:

1. Event manifest and raw MiniSEED under `data/` (when download is run)
2. Phase A.1 EDA — summaries, inspection CSV, four EDA figures
3. Phase A.2 Signal analysis — per-event metrics JSON, eight four-panel figures, summary table
4. Phase B.1 Preprocessing — frozen config, per-event stage NPZ, eight comparison figures, QC table
5. Phase B.2 Feature engineering — **8×18** feature matrix, correlation table, four feature figures

Narrative interpretation lives in `docs/EDA_REPORT.md` through `docs/RESULTS_AND_DISCUSSION.md` and in `reports/Research_Proposal_v1.md`. Those reports **describe** outputs; they do not re-run computations.

The **EarthESND reference implementation** in `src/` is separate (88 tests when the full reference tree is installed). **California pilot results were not produced with EarthESND modules** (`docs/RESULTS_AND_DISCUSSION.md`).

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2. Software environment

Requirement	Source
Python	≥ 3.11 (`pyproject.toml`)
Core packages	ObsPy ≥ 1.5, NumPy, Pandas, Matplotlib, SciPy, PyYAML (`pyproject.toml`)
Optional (reference track only)	pytest ≥ 9.1 (`pyproject.toml [project.optional-dependencies] dev`)

Suggested setup (from repository root):

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -e ".[dev]"
```

There is **no** separate `requirements.txt`; dependencies are declared in `pyproject.toml`.

Network: FDSN download and optional instrument-response steps require internet access to USGS and EarthScope services.

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3. Execution order

Run all commands from the **repository root**. Stages depend on prior outputs as shown.

```
Step 0  Acquisition     → data/raw/iris/, data/manifests/iris_california_pilo
t_events.csv, logs/iris_california_pilot_summary.json
Step 1  EDA (A.1)       → reports/eda/, reports/tables/event_summary.csv, sta
tion_summary.csv, reports/figures/ (4 EDA PNGs)
Step 2  Signal (A.2)    → reports/signal_analysis/, reports/tables/signal_ana
lysis_per_event.csv, reports/figures/ (8 signal PNGs)
Step 3  Preprocess (B.1)→ reports/preprocessing/, reports/tables/preprocessin
g_quality_metrics.csv, reports/figures/ (8 preprocessing PNGs)
Step 4  Features (B.2)  → reports/features/, reports/tables/feature_matrix.cs
v, feature_correlation_matrix.csv, reports/figures/ (4 feature PNGs)
```

Step 0 — Download (optional if raw data already present)

Documented in IRIS_DATASET_REPORT.md §4:

```
python scripts/download/download_iris_california_pilot.py --event-count 8 --m
in-events 5
```

Optional CLI paths: `--raw-dir`, `--manifest-csv` (defaults: `data/raw/iris/`, `data/manifests/iris_california_pilot_events.csv`). The script writes a JSON summary to **logs/iris_california_pilot_summary.json** by default (see scripts/download/download_iris_california_pilot.py).

Public FDSN pilot — no credentials (IRIS_DATASET_REPORT.md §4).

Steps 1–4 — Analysis (manifest-linked traces only)

Commands match phase reports:

```
python scripts/analysis/run_california_pilot_eda.py
python scripts/analysis/run_california_signal_analysis.py
python scripts/analysis/run_california_preprocessing.py
python scripts/analysis/run_california_feature_engineering.py
```

Step	Script	Primary doc
1	scripts/analysis/run_california_pilot_eda.py	EDA_REPORT.md
2	scripts/analysis/run_california_signal_analysis.py	SIGNAL_ANALYSIS_REPORT.md
3	scripts/analysis/run_california_preprocessing.py	PREPROCESSING_REPORT.md
4	scripts/analysis/run_california_feature_engineering.py	FEATURE_ENGINEERING_REPORT.md

Preprocessing parameters for the completed run are recorded in reports/preprocessing/preprocessing_config.json. Feature engineering reads **filtered** arrays from reports/preprocessing/<short_id>_stages.npz and P picks from reports/signal_analysis/<short_id>_metrics.json (FEATURE_ENGINEERING_REPORT.md).

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4. Artefacts: included in the repository vs excluded

4.1 Typically **included** (version-controlled in this project)

These paths are part of the documented pilot and are intended to be shared via Git (see repository commits; audit date 2026-07-29):

Category	Paths
Scripts	scripts/download/download_iris_california_pilot.py, scripts/analysis/run_california_*.py

Category	Paths
Manifest	`data/manifests/iris_california_pilot_events.csv`
Phase documentation	`docs/IRIS_DATASET_REPORT.md`, `docs/EDA_REPORT.md`, `docs/SIGNAL_ANALYSIS_REPORT.md`, `docs/PREPROCESSING_REPORT.md`, `docs/FEATURE_ENGINEERING_REPORT.md`, `docs/RESULTS_AND_DISCUSSION.md`
Tables & summaries	`reports/tables/*.csv`, `reports/eda/*.json`, `reports/eda/mseed_file_inspection.csv`, `reports/signal_analysis/*.json`, `reports/preprocessing/*.json`, `reports/features/*.json`, `reports/features/feature_matrix.csv`
Intermediate arrays	`reports/preprocessing/*_stages.npz`, per-event metrics JSON under `reports/signal_analysis/` and `reports/preprocessing/`
Consolidated narrative	`reports/Research_Proposal_v1.md`

Exact Git tracking may vary by commit; the **manifest and analysis outputs under `reports/` (except figures)** were part of the completed pilot deliverable set per FINAL_DELIVERABLE_AUDIT.md.

4.2 **Intentionally excluded** from Git (`.gitignore`)

Path / pattern	Reason	How to obtain
`data/raw/` (including `data/raw/iris/*.mseed`)	Raw waveforms are large and immutable inputs; re-fetched from FDSN	Step 0 download script
`logs/` (including `logs/iris_california_pilot_summary.json`)	Local run logs	Created when Step 0 runs
`figures/` (matches **`reports/figures/`**)	Generated PNG outputs; reproducible from analysis	Steps 1–4

Path / pattern	Reason	How to obtain
	scripts	
`data/interim/`, **`data/processed/`**, **`data/output/`**	Reserved for other pipelines	Not used for this pilot's published matrix
`.venv/`, **`.env`**	Local environment and secrets	User-created

Audit finding (FINAL_DELIVERABLE_AUDIT.md §1, §5): on a **fresh clone**, **24 PNGs** under `reports/figures/` and **raw MiniSEED** are **not** present unless regenerated or supplied in a separate archive.

4.3 Local-only nuance (documented, not a separate Git policy)

EDA_REPORT.md and `reports/eda/mseed_file_inspection.csv` record **twelve** MiniSEED files on disk at EDA time (**eight** manifest-linked + **four legacy** from an earlier download). **All completed analyses use the eight manifest-linked records only.**

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5. Regenerating excluded artefacts

Excluded artefact	Regeneration
Raw MiniSEED	`python scripts/download/download_iris_california_pilot.py --event-count 8 --min-events 5`
Acquisition JSON log	Same download run → `logs/iris_california_pilot_summary.json`
EDA figures (4)	`python scripts/analysis/run_california_pilot_eda.py`
Signal figures (8)	`python scripts/analysis/run_california_signal_analysis.py`
Preprocessing figures (8)	`python scripts/analysis/run_california_preprocessing.py` (requires prior signal metrics for consistency with documented chain)
Feature figures (4)	`python scripts/analysis/run_california_feature_engineering.py` (requires preprocessing NPZ + signal metrics)

Full figure index: `reports/Research_Proposal_v1.md` Appendix A; filenames listed in phase reports.

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6. Reproducibility limitations

These limits are **documented project findings**, not new science:

1. **FDSN variability** — Re-running Step 0 may yield different events or stations if catalog or dataselect availability changes; the **published manifest** (8 events, CI.ADO / CI.USC) reflects the **2026-07-29** run (IRIS_DATASET_REPORT.md §3).

2. **Station selection** — Fixed distance-ranked candidate list; not uniform network sampling (IRIS_DATASET_REPORT.md §2.2, §5).

3. **Origin-aligned windows** — 300 s from catalog origin; no pre-event quiet time for noise (SIGNAL_ANALYSIS_REPORT.md §1.2).

4. **Instrument response** — Preprocessing run: response removal **0/8 applied**; filtered traces remain in **counts** (PREPROCESSING_REPORT.md §4).

5. **Small sample** — **N = 8** events, single **BHZ** component; statistics are illustrative (RESULTS_AND_DISCUSSION.md).

6. **EarthESND** — Reference code in `src/` is **not** on the California reproduction path.

7. **Bit-identical PNGs** — Matplotlib/ObsPy versions may produce minor visual differences; tabular/json outputs should match for the same inputs and manifest.

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7. Verification pointers

After regeneration, compare against frozen summaries (completed pilot):

- `reports/eda/dataset_summary.json`

- `reports/signal_analysis/signal_analysis_summary.json`

- `reports/preprocessing/preprocessing_summary.json`

- `reports/features/feature_summary.json`

Phase reports and FINAL_DELIVERABLE_AUDIT.md list expected file counts (e.g. **24** PNGs, **8** manifest events).

8. References (software and methods cited in repository)

Krischer, L., Megies, T., Barsch, R., Beyreuther, M., Lecocq, T., Caudron, C., and Wassermann, J. (2015). ObsPy: A bridge for earthquake science. *Seismological Research Letters*, 86(3), 765–771.

International Federation of Digital Seismograph Networks (FDSN). FDSN web services specification. https://www.fdsn.org/webservices/

Project documentation: IRIS_DATASET_REPORT.md, phase analysis reports under `docs/`, Research_Proposal_v1.md §12.

Document control: D-S1 reproducibility statement v1.0 — 29 July 2026. No source code modified to produce this file.